

Storage Class, Server Access Logging and Policy Generator

Class no:7

What is a Storage Class in AWS S3?

- A **storage class** in Amazon S3 specifies the **cost, availability, durability, and retrieval time** of objects stored in buckets.(as they can be applied only on object level)
- It allows you to optimize storage costs by choosing the right class based on **data access patterns** (frequent, infrequent, archival). They are applicable for both private and public buckets.

Purpose of Storage Classes

- **Cost optimization:** Pay less for data that is rarely accessed.
- **Performance tuning:** Ensure fast retrieval for frequently accessed data.
- **Data lifecycle management:** Move data automatically between classes as access patterns change.
- **Compliance & durability:** Meet business requirements for availability, redundancy, and geographic isolation.

Types of AWS S3 Storage Classes and Their Purpose

Storage Class	Durability & Availability	Cost	Purpose
S3 Standard	99.999999999% durability, high availability	Higher	For frequently accessed data (websites, apps, analytics).
S3 Express One Zone	High performance, single AZ	Lower than Standard	Ultra-low latency for frequently accessed data in one zone.
S3 Intelligent-Tiering	Same durability as Standard	Variable	Automatically moves data between frequent/infrequent tiers based on usage. Best for unknown or changing access patterns.
S3 Standard-IA (Infrequent Access)	High durability, lower availability	Lower	For data accessed less often but needs quick retrieval (backups, disaster recovery).
S3 One Zone-IA	Stored in one AZ only	Lower	For infrequently accessed data that can be re-created if lost.
S3 Glacier Instant Retrieval	Millisecond retrieval	Very low	For archival data that still needs occasional instant access.
S3 Glacier Flexible Retrieval	Minutes to hours retrieval	Very low	For archives where retrieval speed is flexible.
S3 Glacier Deep Archive	Hours retrieval	Lowest	For long-term archival (regulatory, compliance, rarely accessed data).

Key Points

- **Standard** → Best for hot data (frequent use).
- **Intelligent-Tiering** → Best when access patterns are unpredictable.
- **IA classes** → Balance cost and speed for backups or rarely used files.
- **Glacier classes** → Lowest-cost archival storage, retrieval times vary.
- **Express One Zone** → High-speed, single-zone option for performance-sensitive workloads.

Hands on: Creating a bucket and implementing the storage class on object.

Step no:1 Create a bucket : `s3bucketforstorageclassdemo12345` (it can be either private or public). Here we are creating a private bucket.

Buckets

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (1) [Info](#)

Buckets are containers for data stored in S3.

< 1 > [Settings](#)

	Name	AWS Region	Creation date
☐	s3bucketforstorageclassdemo12345	Europe (Stockholm) eu-north-1	March 17, 2026, 19:20:34 (UTC+05:30)

Step no:2 Now try to add objects in the bucket. In the page we can see the **properties** section, click on that we can decide the type of storage class depends on the object, how often we are going to access it. Then choose **upload**.

Upload [Info](#)

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (1 total, 119.1 KB)

All files and folders in this table will be uploaded.

☐	Name	Folder	Type	Size
☐	v.jpg	-	image/jpeg	119.1 KB

Destination [Info](#)

Destination

[s3://s3bucketforstorageclassdemo12345](#)

► Destination details

Bucket settings that impact new objects stored in the specified destination.

► Permissions

Grant public access and access to other AWS accounts.

▼ Properties

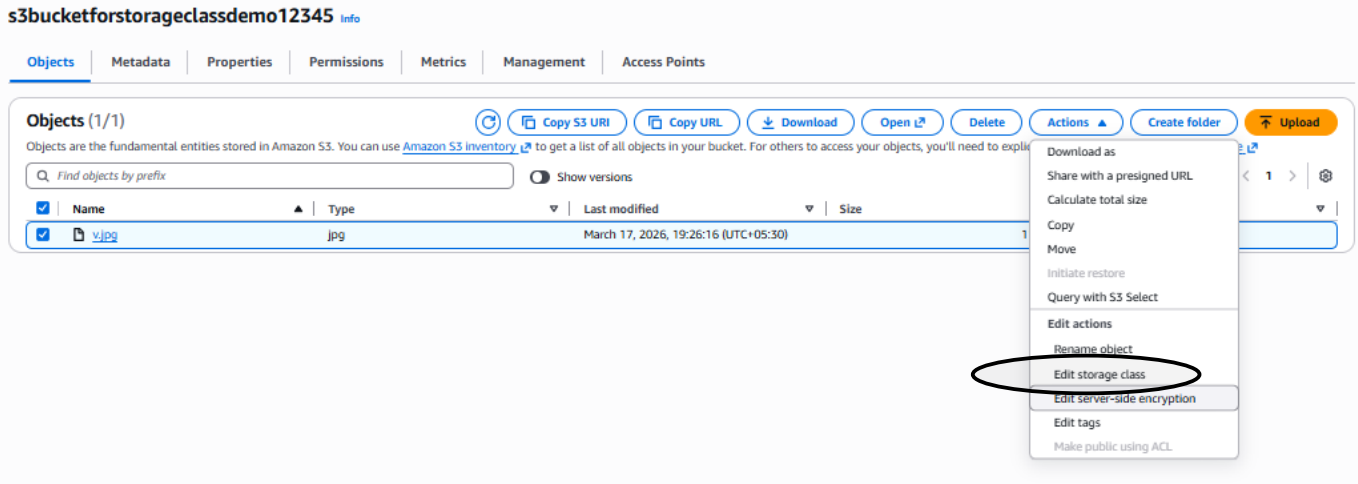
Specify storage class, encryption settings, tags, and more.

Storage class [Info](#)

Amazon S3 offers a range of storage classes designed for different use cases. [Learn more](#) or see [Amazon S3 pricing](#)

	Storage class	Designed for	Bucket type	Availability Zones	Min storage duration	Min billable object size	Monitoring and auto-tiering fees	Retrieval fees
☒	Standard	Frequently accessed data (more than once a month) with milliseconds access	General purpose	≥ 3	-	-	-	-
☐	Intelligent-Tiering	Data with changing or unknown access patterns	General purpose	≥ 3	-	-	Per-object fees apply for objects >= 128 KB	-
☐	Standard-IA	Infrequently accessed data (once a month) with milliseconds access	General purpose	≥ 3	30 days	128 KB	-	Per-GB fees apply
☐	One Zone-IA	Recreatable, infrequently accessed data (once a month) with milliseconds access	General purpose or directory	1	30 days	128 KB	-	Per-GB fees apply
☐	Glacier Instant Retrieval	Long-lived archive data accessed once a quarter with instant retrieval in milliseconds	General purpose	≥ 3	90 days	128 KB	-	Per-GB fees apply
☐	Glacier Flexible Retrieval (formerly Glacier)	Long-lived archive data accessed once a year with retrieval of minutes to hours	General purpose	≥ 3	90 days	-	-	Per-GB fees apply
☐	Glacier Deep	Long-lived archive data accessed less than	General	> 1	180 days	-	-	Per-GB fees apply

Also, we can change the storage class of an object after uploading it. Click on the object, choose **actions** where we can see the **Edit storage class option**



Lifecycle rule in S3: (Storage class setup)

- A **Lifecycle rule** in Amazon S3 is a policy that automatically manages the **transition** and **expiration** of objects in a bucket.
- It defines actions to move objects between **storage classes** or delete them after a certain period.

Purpose of Lifecycle Rules

- **Cost optimization:** Move data from expensive storage classes (like Standard) to cheaper ones (like Glacier) when it becomes less frequently accessed.
- **Automation:** Reduce manual effort by automatically handling data transitions and deletions.
- **Data management:** Ensure compliance with retention policies by expiring objects after a set time.
- **Efficiency:** Keep buckets clean and avoid unnecessary storage costs.

Key Actions in Lifecycle Rules

1. Transition Actions

- Move objects to another storage class after a specified number of days.
- Example: Move logs from **S3 Standard** → **S3 Standard-IA** after 30 days → **S3 Glacier Deep Archive** after 1 year.

2. Expiration Actions

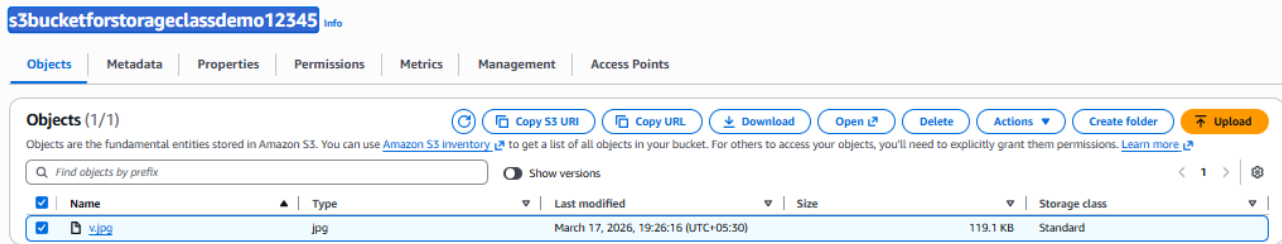
- Permanently delete objects after a defined period.
- Example: Delete temporary files after 7 days.

3. Abort Incomplete Multipart Uploads

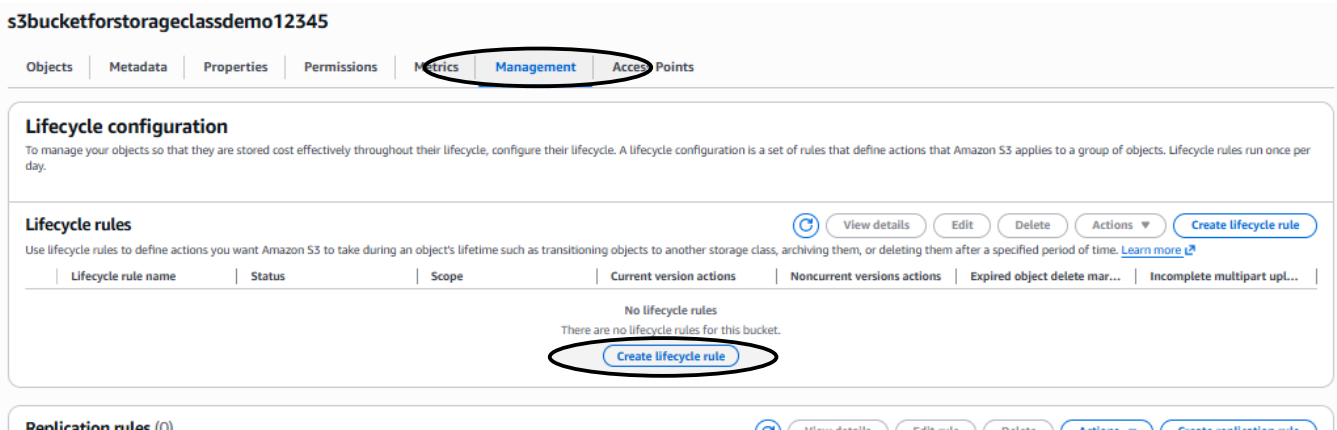
- Automatically clean up incomplete uploads after a set time to save costs.

Hands on for lifecycle rule set up in our created S3 bucket.

Step no:1 Open our bucket (s3bucketforstorageclassdemo12345) -> move to the object.



Step no:2 Click on the **management** tab on the object page and go to **create lifecycle rule** click on it.



Step no:3 Provide a rule name, enable apply to all objects, enable transition for current versions, acknowledge it.

Create lifecycle rule Info

Lifecycle rule configuration

Lifecycle rule name

Noeruleforstorageclassdemo12345

Up to 255 characters

Choose a rule scope

- ☐ Limit the scope of this rule using one or more filters
- ☒ Apply to all objects in the bucket

⚠ Apply to all objects in the bucket

If you want the rule to apply to specific objects, you must use a filter to identify those objects. Choose "Limit the scope of this rule using one or more filters". [Learn more](#)

☒ I acknowledge that this rule will apply to all objects in the bucket.

Lifecycle rule actions

Choose the actions you want this rule to perform.

- ☒ Transition current versions of objects between storage classes
This action will move current versions.
- ☐ Transition noncurrent versions of objects between storage classes
This action will move noncurrent versions.
- ☐ Expire current versions of objects
- ☐ Permanently delete noncurrent versions of objects
- ☐ Delete expired object delete markers or incomplete multipart uploads
These actions are not supported when filtering by object tags or object size.

⚠ Transitions are charged per request

For a lifecycle transition action, each request corresponds to an object transition. For details on lifecycle transition pricing, see requests pricing info on the [Storage & requests](#) tab of the [Amazon S3 pricing page](#).

☒ I acknowledge that this lifecycle rule will incur a transition cost per request.

ⓘ By default, objects less than 128KB will not transition across any storage class

We don't recommend transitioning objects less than 128 KB because the transition costs can outweigh the storage savings. If your use case requires transitioning objects less than 128 KB, specify a minimum object size filter for each applicable lifecycle rule with a transition action.

Now we will configure the **Transition current versions of objects between storage classes**. This is on the same page. Click create rule. It will automate the process.

Note: always check and enter the minimum days for the storage class objects.

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions	Days after object creation	
Standard-IA	30	<button>Remove</button>
Intelligent-Tiering	60	<button>Remove</button>
One Zone-IA	95	<button>Remove</button>
Glacier Flexible Retrieval (formerly Glacier)	190	<button>Remove</button>

Add transition

Review transition and expiration actions

Current version actions

Day 0

- Objects uploaded

↓

Day 30

- Objects move to Standard-IA

↓

Day 60

- Objects move to Intelligent-Tiering

↓

Day 95

- Objects move to One Zone-IA

↓

Day 190

- Objects move to Glacier Flexible Retrieval (formerly Glacier)

Noncurrent versions actions

Day 0

No actions defined.

Cancel Create rule

Server access Logging:

- Server Access Logging is a feature in Amazon S3 that records detailed information about requests made to your S3 bucket.
- It generates log files that capture details such as the requester, bucket name, request time, action performed, error codes, and more.
- Logs are stored in a designated S3 bucket for later analysis.

Purpose of Server Access Logging

- Audit & Compliance: Track who accessed data and when, useful for security reviews.
- Security Monitoring: Detect unauthorized or suspicious access attempts.
- Usage Analysis: Understand access patterns, frequency, and performance.
- Cost Optimization: Identify unnecessary requests or inefficient data access.
- Troubleshooting: Diagnose issues related to permissions or failed requests.

Key Aspects of Server Access Logging

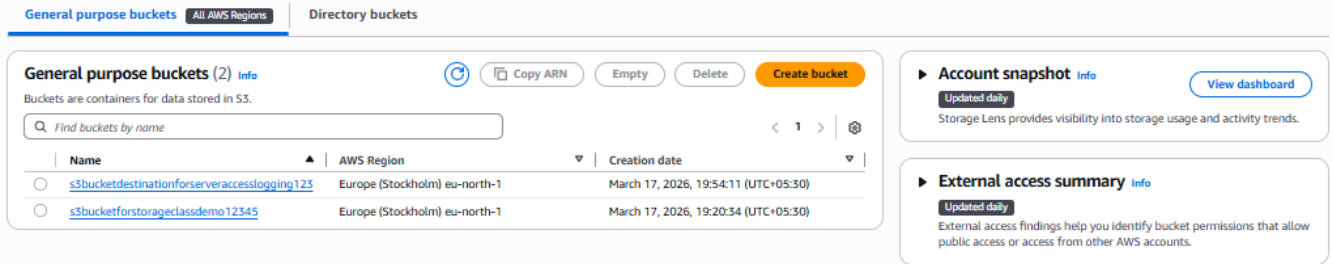
Log Destination and Log Content

- Logs are delivered to a target S3 bucket (must be in the same AWS region).
- Best practice: use a separate bucket for logs to avoid clutter.

Hands on for Creating a server access logging:

Step no:1 Create two buckets. One is our bucket we want to monitor (source:s3bucketforstorageclassdemo12345) and the other (destination: s3bucketdestinationforserveraccesslogging123) is where we store the logs of the source bucket.

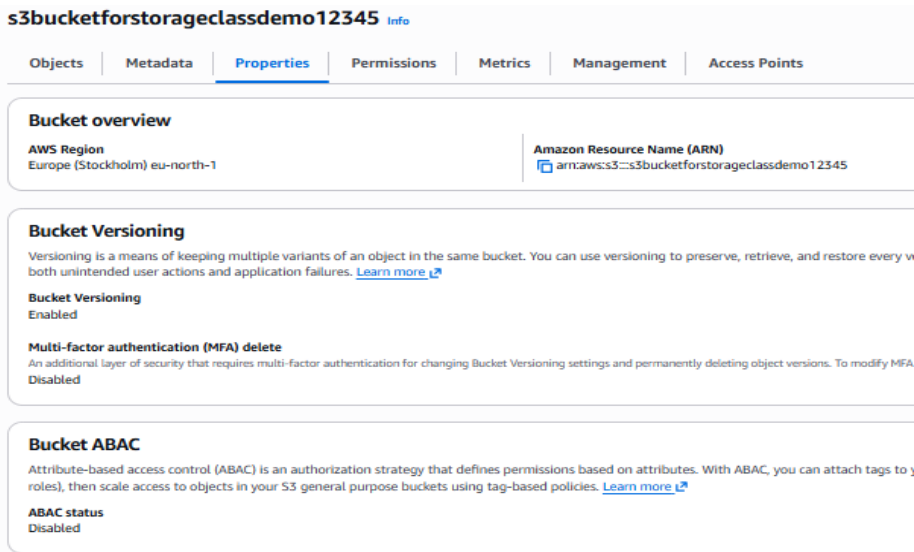
Buckets



The screenshot shows the AWS S3 Buckets console. The 'General purpose buckets' tab is selected, showing a list of two buckets. The first bucket is 's3bucketdestinationforserveraccesslogging123' and the second is 's3bucketforstorageclassdemo12345'. Both are in the 'Europe (Stockholm) eu-north-1' region. To the right, there are two panels: 'Account snapshot' and 'External access summary', both with 'Updated daily' status and 'View dashboard' links.

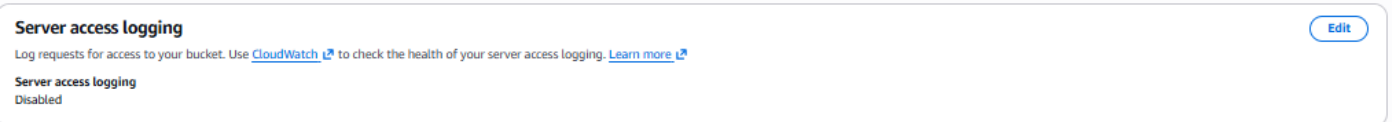
Name	AWS Region	Creation date
s3bucketdestinationforserveraccesslogging123	Europe (Stockholm) eu-north-1	March 17, 2026, 19:54:11 (UTC+05:30)
s3bucketforstorageclassdemo12345	Europe (Stockholm) eu-north-1	March 17, 2026, 19:20:34 (UTC+05:30)

Step no:2 Open our source bucket we want to monitor. Click the properties tab.



The screenshot shows the 'Properties' tab of the 's3bucketforstorageclassdemo12345' bucket. The 'Bucket overview' section shows the AWS Region as 'Europe (Stockholm) eu-north-1' and the Amazon Resource Name (ARN) as 'arn:aws:s3::s3bucketforstorageclassdemo12345'. The 'Bucket Versioning' section shows 'Bucket Versioning' as 'Enabled' and 'Multi-factor authentication (MFA) delete' as 'Disabled'. The 'Bucket ABAC' section shows 'ABAC status' as 'Disabled'.

Scroll down. We can see the **server access logging** option. Click **edit**.



The screenshot shows the 'Server access logging' settings page. The 'Server access logging' status is 'Disabled'. There is an 'Edit' button in the top right corner. The text below the status says: 'Log requests for access to your bucket. Use [CloudWatch](#) to check the health of your server access logging. [Learn more](#)'.

Step no:3 Enable Server access login.

- Provide the destination bucket name, followed by the suffix /logs to easily identify as log files.
- We can choose the desired format based on that it will show an example below.
- Save changes.

Wait at least 30 minutes to see the logs in the destination bucket we have provided for the changes in the source bucket.

Edit server access logging [Info](#)

Server access logging

Log requests for access to your bucket. [Learn more](#)

Server access logging

- ☐ Disable
☒ Enable

⚠ Bucket policy will be updated

When you enable server access logging, the S3 console automatically updates your bucket policy to include access to the S3 log delivery group.

Destination

Specify a destination bucket in the Europe (Stockholm) eu-north-1 Region. To store your logs under a particular prefix, make sure that you include a slash (/) after the name of the prefix. Otherwise, the prefix will be added to the name of your log files.

s3://s3bucketdestinationforserveraccesslogging123/logs

[Browse S3](#)

Format: s3://<bucket>/<optional-prefix-with-path>

Destination Region

Europe (Stockholm) eu-north-1

Destination bucket name

s3bucketdestinationforserveraccesslogging123

Destination prefix

logs

Log object key format

- ☐ [DestinationPrefix][YYYY-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]]
☒ [DestinationPrefix][SourceAccountId]/[SourceRegion]/[SourceBucket]/[YYYY]/[MM]/[DD]/[YYYY]-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]
To speed up analytics and query applications, use this format.

Source of date used in log object key format

- ☐ S3 event time
The year, month, and day will be based on the timestamp of the S3 event in the file that's been delivered.
☒ Log file delivery time
The year, month, and day will be based on the time when the log file was delivered to S3.

Log object key example

logs314383175192/eu-north-1/s3bucketforstorageclassdemo12345/2026/07/01/2026-07-01-10-12-56-[UniqueString]

[Cancel](#)

[Save changes](#)

Policy Generator:

- A **web-based tool** that assists in creating **access control policies** for AWS resources.
- It generates **JSON policy documents** that define what actions are allowed or denied on specific AWS services.
- Supports multiple policy types: **IAM policies**, **S3 bucket policies**, **SNS topic policies**, **SQS queue policies**, **VPC endpoint policies**, etc.

Purpose of the Policy Generator

- **Ease of Use:** Helps users who are not familiar with JSON syntax create valid policies.
- **Consistency:** Ensures policies follow AWS best practices and correct structure.
- **Security:** Allows fine-grained control over access permissions, reducing misconfigurations.
- **Efficiency:** Saves time compared to writing policies manually.

Hands on: Creating a policy for denying an IAM user to access a specific s3 bucket, even though the user has S3 full access policy.

(Note: The user cannot access the specific bucket alone, except that rest of the buckets can be accessed by the user.)

Step no:1 Create an IAM user with S3 full access policy.

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name: Nithyas3FullAccess
Console password type: Custom password
Require password reset: No

Permissions summary

Name: AmazonS3FullAccess
Type: AWS managed
Used as: Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.
No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#)

[Previous](#)

[Create user](#)

Step no:2 Search AWS policy generator for S3 in the browser and open it.



awspolicygen.s3.amazonaws.com

<https://awspolicygen.s3.amazonaws.com/policygen.html>

AWS Policy Generator

This AWS Policy Generator is provided for informational purposes only, you are still responsible for your use of Amazon Web Services technologies and ensuring that your use is in compliance with all ...

Step no:4 Follow the configurations,

- Type of Policy: S3 Bucket Policy
- Deny
- ARN of the User
- Enable All actions
- ARN of the bucket
- Add statement
- Generate policy

Statements added (1)

You added the following statements. Click the button below to Generate a policy.

Principal(s)	Effect	Action	Resource(s)	Condition(s)	Remove
arn:aws:iam::314383175192:user/Nithyas3fullaccess	Deny	s3:*	arn:aws:s3:::s3bucketforstorageclassdemo12345	None	Remove

Step 3: Generate policy

A policy is a document (written in the [Access Policy Language](#)) that acts as a container for one or more statements.

[Generate Policy](#)

Copy the policy contents.

Step no:3 Open our bucket in the console, move to permissions. We can see the bucket policy option. Choose edit.

s3bucketforstorageclassdemo12345 [Info](#)

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [Access Points](#)

Permissions overview

Access finding

Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#).
[View analyzer for eu-north-1](#)

Block public access (bucket settings) [Edit](#)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access

[On](#)

► Individual Block Public Access settings for this bucket

Bucket policy [Edit](#) [Delete](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

❗ Public access is blocked because Block Public Access settings are turned on for this bucket

To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about [using Amazon S3 Block Public Access](#)

Step no:4 Now paste the contents of the policy generator. It will be in json format. Save changes.

Edit bucket policy [Info](#)

Bucket policy

[Policy examples](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Bucket ARN

arn:aws:s3::s3bucketforstorageclassdemo12345

Policy

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Sid": "Statement1",  
6       "Effect": "Deny",  
7       "Principal": {  
8         "AWS": "arn:aws:iam::314383175192:user/Nithyas3fullaccess"  
9       },  
10      "Action": "s3:*",  
11      "Resource": "arn:aws:s3::s3bucketforstorageclassdemo12345"  
12    }  
13  ]  
14 }
```

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

[+ Add new statement](#)[+ Add new statement](#)

JSON Ln 14, Col 1

Security: 0 Errors: 0 Warnings: 0 Suggestions: 0

[Preview external access](#)[Cancel](#)[Save changes](#)

Step no:5 To check: Open a private window and login as the IAM user who cannot access this bucket.

IAM user sign in [i](#)

Account ID or alias [\(Don't have?\)](#)

314383175192

☐ Remember this account

IAM username

Nithyas3fullaccess

Password

••••••••



☐ Show Password

[Having trouble?](#)

[Sign in](#)

Then try to access the specific bucket. It will show access denied. It is because we have enabled all actions to be denied during the policy generation.

s3bucketforstorageclassdemo12345Info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects

Copy S3 URI

Copy URL

Download

Open ↗

Delete

Actions ▼

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

< 1 >

NameTypeLast modifiedSizeStorage class

ⓘ Insufficient permissions to list objects

After you or your AWS administrator has updated your permissions to allow the s3:ListBucket action, refresh the page. Learn more about [Identity and access management in Amazon S3](#)

Diagnose with Amazon Q